Question 1 **5 marks**

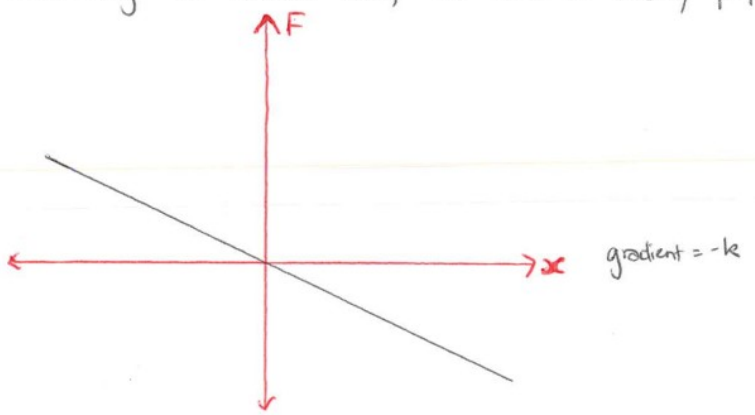
(a) ***Hooke's Law***

(i) State Hooke's Law as an equation, and explain the equation in words.

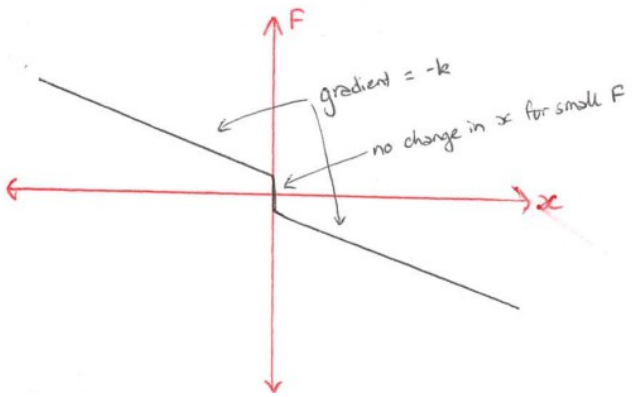
F=-k**x**. The force of an extended (compressed) spring is opposed to the extension (compression) and linearly proportional to the magnitude of the extension (compression) and in the same line. The proportionality constant is the spring constant k.

(ii) Assume that this equation relates to a single one-dimensional spring. Explain how this law might need re-evaluating for small forces in the presence of static friction. Draw a plot of *F* versus *x* according to Hooke's Law, and then draw a plot of *F* versus *x* of what the shape might be in the presence of static friction.

ii) According to Hooke's Law, the force is linearly proportional always.



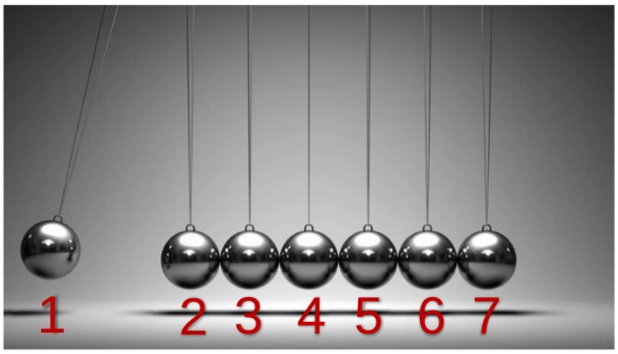
When including static friction, small forces (that are less than the max static friction force) will produce no displacement. So, for small values of F, the line will be translated:



[2 + 3 = 5 marks]

Question 2 **10 marks**

Consider the figure, to the right, of a Newton's cradle which has 7 balls, which for the moment can be considered to be of equal mass. For the questions below assume that the initial state is defined by the impact velocity of ball 1 or balls 1 and 2 on the remaining stationary balls and the final state is the velocity of ball 7 or balls 6 and 7 immediately after the collision.



(a) If ball 1 hits the remaining stationary balls at a velocity v show that the in the final state ball 7 must have a velocity of v.

(b) If balls 1 and 2 hit the remaining, stationary, balls at a velocity v show that in the final state balls 6 and 7 must have a velocity of v.

(c) Now consider the case where masses of balls 1, 2, 3, 4, 5, 6 and 7 are m_1 = m, m_2 = 2m, m_3 = 3m, m_4 = 4m, m_5 = 3m, m_6 = m and m_7 = 2m respectively.

(i) If balls 1 and 2 hit the remaining, stationary, balls at a velocity v show that in the final state balls 6 and 7 must have a velocity of v.

(ii) If ball 1 hits the remaining, stationary balls, at a velocity v is it possible to calculate the final state?

[2 + 3 + (3 + 2) = 10 marks]

(a) Conservation of momentum: $m v_1 = k m v_2$ where k is the number of balls.

Conservation of energy: $\frac{1}{2} m v_1^2 = \frac{1}{2} k m v_2^2 = \frac{1}{2} k m \left(\frac{m^2 v_1^2}{k^2 m^2} \right)$ only true if k=1.

So only one ball leaves, and must have velocity $v_2 = v_1 = v$.

(b) Likewise, $2 m v_1 = k m v_2$, so $2 v_1 = k v_2$.

$\frac{1}{2} 2 m v_1^2 = \frac{1}{2} k m v_2^2 = \frac{1}{2} k m \left(\frac{2 v_1}{k} \right)^2$, so $m v_1^2 = \frac{2 m v_1^2}{k}$, and therefore k = 2 (so there are two balls), and since $2 v_1 = k v_2$, they each have speed $v_2 = v_1 = v$.

(c) (i) Again, by conservation of momentum, $3 m v_1 = k m v_2$, so $v_2 = \frac{3 v_1}{k}$.

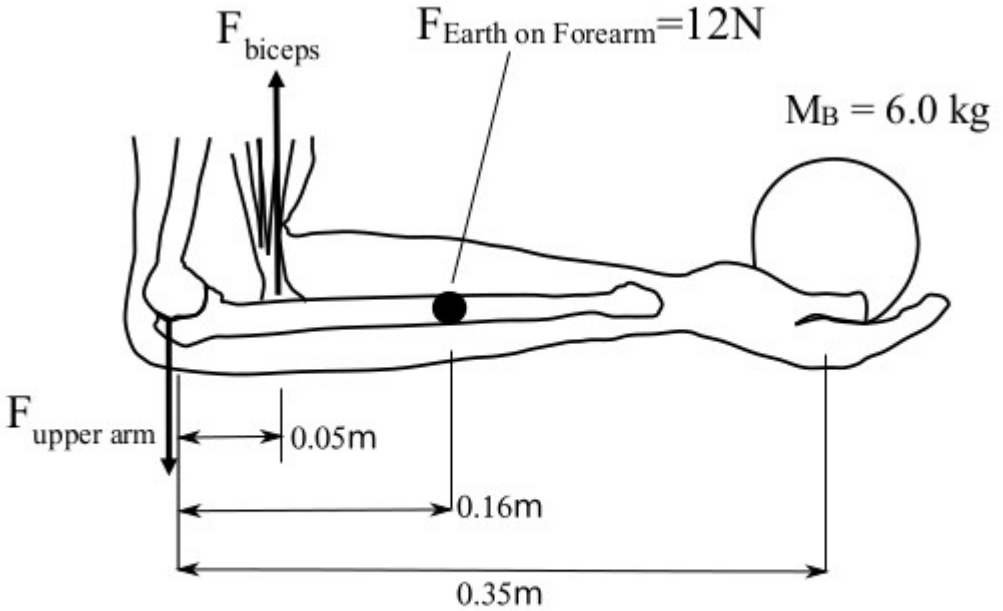
By conservation of energy, $\frac{1}{2} 3 m v_1^2 = \frac{1}{2} k m v_2^2 = \frac{1}{2} k m \left(\frac{9 v_1^2}{k^2} \right)$ so $3 v_1^2 = \frac{9 v_1^2}{k}$.

So k = 9/3 = 3. So the total mass of the final state is 3m, which is ball 6 + ball 7, each with speed v.

(ii) No it is not possible by just considering the velocity of just ball 6 & 7 in the final state, as we can not conserve energy and momentum. To calculate the final state, we need to consider the momentum and energy of all the balls.

Question 3 **18 marks**

A 6.0 kg ball is held stationary in the hand of a person with their arm bent as shown in the figure below. The arm pivots at the elbow joint as shown. The arm is supported by the biceps muscle which is attached 5.0 cm from the elbow. The weight force of the arm is 12 N and is directed through the centre of mass of the arm located 0.16m from the elbow joint. The weight of the ball is applied 0.35m from the elbow.



A 6.0 kg ball held stationary in a bent arm as shown. Assume all the forces are perpendicular to the arm at the locations shown.

a) Calculate the torque on the elbow from the weight force of the arm.
b) Calculate the torque on the elbow from the weight of the ball, assume $g = 9.8\,ms^{-2}$.
c) Use the principle of static equilibrium to determine the force from the biceps on the arm.
d) Use the principle of static equilibrium to determine the force from the upper arm.
e) If the bicep has a maximum force limit of 1500 N, what would be the largest mass the arm could hold?

[3 + 4 + 4 + 3 + 4 = 18 marks]

(a) Torque due to weight of arm = $12 \times 0.16 = 1.92\,Nm$

(b) Torque due to weight of ball = $6.0 \times 9.8 \times 0.35 = 20.6\,Nm$

(c) Static equilibrium => Torques must sum to zero

$-\tau_{biceps} + \tau_{arm} + \tau_{Ball} = 0$

$\Rightarrow \tau_{biceps} = 1.92 + 20.6 = 22.5\,Nm$

- ⇨ Force of biceps = $\frac{22.5}{0.05} = 450\,N$

(d) Static equilibrium => All forces must sum to zero

$-F_{upper\,arm} + F_{biceps} - F_{arm} - F_{ball} = 0$

$\Rightarrow F_{upper\,arm} = F_{biceps} - F_{arm} - F_{ball}$

$= 450 - 12 - 6.0 \times 9.8 = 379\,N$

(e) If max force of biceps is 1500 N => Max Torque of biceps $\tau_{biceps} = 0.05 \times 1500 = 75\,Nm$

Static equilibrium => Torques must sum to zero

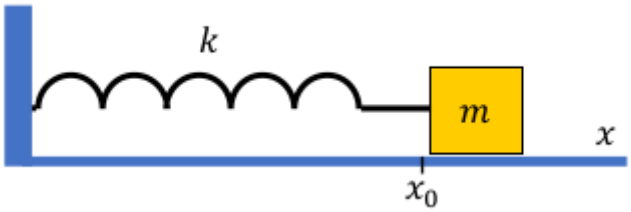
$-\tau_{biceps} + \tau_{arm} + \tau_{Ball} = 0$

- ⇨ $\tau_{Ball} = \tau_{biceps} - \tau_{arm} = 75 - 1.92 = 73.1\,Nm$
- ⇨ Larges Mass of Ball = $\frac{73.1}{0.35 \times 9.8} = 21\,kg$

Question 4 **25 marks**

A mass $m = 2.00\,kg$ fixed to a spring with spring constant $k = 280\,N/m$ is sliding on a frictionless surface (see figure), oscillating according to the equation of motion $x(t) = A\sin\omega t$.

(a) Calculate the frequency and period of oscillation.
(b) Calculate the length of a pendulum with the same mass and period of oscillation.
(c) The maximum potential energy of the system is $U = 0.70\,J$. Determine the value of A.
(d) Plot the displacement of the mass against time, for one complete cycle of oscillation, with correct scales.
(e) Derive an expression for the velocity of the mass.
(f) Calculate the maximum speed of the mass.
(g) Plot the velocity of the mass against time, for one complete oscillation, with correct scales



[4 + 3 + 3 + 5 + 4 + 2 + 4 = 25 marks]

(a) $\omega = \sqrt{\frac{k}{m}}$ = 11.8 rad/s = 1.89 Hz

$T = \frac{2\pi}{\omega}$ = 0.53 s

(b) $T_{pendulum} = 2\pi\sqrt{\frac{L}{g}}$; $L = \left(\frac{T}{2\pi}\right)^2 g$ = 70 mm

(c) $U_{max} = \frac{1}{2}k\,A^2$; $A = \sqrt{\frac{2U}{k}}$ = 0.071m = 71 mm

(d) Sine curve with one full cycle ; Correctly labelled axes ; Correct amplitude on y axis ; Correct period on x axis

(e) $v = \frac{dx}{dt} = \frac{d}{dt}\left[A\sin(\omega t)\right] = A\omega\cos(\omega t) = 0.83\cos(11.8t)$

(f) $v_{max} = A\omega = 0.83\,m/s$

(g) Correct axis labels and units ; Correct shape of cosine